

Understanding Sentiment in Online Music Communities Through Batch and Real-Time Analysis

Abstract

Online platforms have become powerful spaces where music listeners express preferences, negotiate identity, and participate in cultural dialogue. Among these platforms, YouTube and Reddit serve as high-traffic hubs where users discuss genres, artists, releases, controversies and trends. As computational methods advance, sentiment analysis provides new tools for quantifying emotional expression within these communities. This study develops an integrated system that combines a batch-processing pipeline for the analysis of large volumes of historical YouTube comments with a real-time streaming architecture that continuously monitors sentiment in Reddit music communities. Using VADER and TextBlob to evaluate emotional tone, the project explores how sentiment varies across genres (Hip Hop, Indie, Pop) and high-profile artists (Drake, Kendrick Lamar).

Batch data was collected using genre-specific and artist-specific queries, then processed through a PySpark pipeline for cleaning, language filtering, sentiment scoring, and classification. Streaming data was captured using PRAW, stored in MongoDB, and visualised in real time using rolling averages within an interactive dashboard. Results indicate that Indie and Pop comments show the most consistent positivity, while Hip Hop comments tend to be more polarised. Artist analysis suggests that Drake commentary is more emotionally extreme and volatile, whereas Kendrick Lamar commentary is steadier and far more positive overall. Real-time data further reveals temporal fluctuations, especially within HipHopHeads.

By integrating scalable engineering techniques with cultural analysis, this work demonstrates how computational sentiment analysis can offer insights into the dynamics of online music communities.

1. Introduction

Music has long played a central role in shaping individual and collective identities, and online platforms have amplified the speed and scale at which these identities are expressed. From YouTube reaction videos to Reddit genre communities, social media has become a thriving ecosystem for music commentary. These platforms not only facilitate fan participation but also influence artist reputation, genre perception, and even commercial success. As a result, understanding the emotional tone of discourse in these environments is of increasing interest to researchers, industry professionals, and cultural analysts.

The aim of this study is to investigate how sentiment manifests within online music communities, using a dual framework that integrates batch processing of historical YouTube comments using PySpark, and real-time streaming of Reddit comments from three active music subreddits.

This dual approach enables both macro-level and micro-level insights. Batch analysis captures aggregated sentiment patterns across thousands of comments collected over time. Streaming analysis captures sentiment as it evolves minute-by-minute, illustrating how online communities respond to live discussions, breaking news, and cultural events.

The project focuses on three genres: Hip Hop, Indie and Pop, as well as two major artists: Drake and Kendrick Lamar. These selections represent contrasting musical identities and fan communities. Hip Hop communities often feature lyrical critique, competitive discourse, and cultural commentary. Indie communities tend to value emotion, authenticity, and artistic depth. Pop communities feature intense fan engagement and celebrity-driven narratives. Drake and Kendrick Lamar offer an additional layer of comparison: both are globally influential, but their public personas and fan communities differ significantly.

This report addresses the following research questions:

- How does sentiment vary across genres in a large corpus of YouTube comments?
- How do online audiences express sentiment toward Drake and Kendrick Lamar?
- How does real-time sentiment differ from aggregated batch sentiment?
- What methodological and ethical considerations arise when analysing social media data?

2. Description of the Project and Topic

The overarching purpose of this project is to examine emotional expression within online music communities by analysing user-generated comments. The project incorporates:

- Batch processing for large-scale, historical data.
- Real-time streaming for live insights.
- Sentiment analysis using two different algorithms.
- Genre and artist classification.
- Dashboard visualisation.

The topic was chosen because music communities offer rich emotional and cultural dialogue, making them ideal for computational analysis. Social media provides massive, continuously evolving datasets, which allows for both retrospective and real-time investigation.

The dual-architecture approach differentiates this project from studies relying solely on static datasets or single-platform analysis. Combining YouTube and Reddit enables cross-platform comparison and demonstrates how user behaviour differs across contexts.

3. Batch Data Collection: Method and Considerations

Batch data was collected using the YouTube Data API v3. A systematic query strategy was designed to ensure diversity and avoid bias toward isolated videos. Three key design principles informed the collection process:

3.1 Query Breadth

Multiple search terms were defined for each genre:

- Hip Hop: *hip hop music, rap music, 90s rap*

- Indie: *indie music, indie rock, indie pop*
- Pop: *pop music, billboard pop, top pop hits*

Artist-specific queries included interview, reaction, performance, and album review searches for both Drake and Kendrick Lamar, as well as the song titles of their diss tracks towards each other. “Drake vs Kendrick” queries captured discourse surrounding rivalry and comparison.

3.2 Metadata Enrichment

Each retrieved comment was tagged with additional metadata:

- genre_hint (from query).
- artist_hint (for artist-specific searches).
- Video ID.
- Author name.
- Timestamp.

This contextual metadata supported downstream classification.

3.3 Ethical and Practical Considerations

Several considerations shaped the design:

- YouTube API quota limits restricted volume, requiring careful query selection.
- Users do not explicitly consent to data being used for research, so identifiable content was excluded from the report.
- YouTube’s algorithm prioritises trending or highly engaged content, meaning results reflect platform bias.
- Not all videos allow comments; some queries returned no usable data.

Despite these challenges, the collection process yielded several thousand usable comments.

4. Random Sample of Collected Batch Posts

A random selection of posts was manually reviewed to validate data quality. The samples include emotionally charged reactions, general music opinions, and varied linguistic patterns. Typical examples (with usernames removed) include:

“This track is unbelievable, Drake never misses.”

“Kendrick’s storytelling gives me chills every time.”

“I love indie music but this mix sounds flat.”

“Pop songs are so repetitive lately.”

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.....
|She could be a transformer. He wasn't taking no risk.
|she could be a transformer he wasnt taking no risk
|
|Im shocked Tracklib took this long to put this out. Its just the Al Green track with looped vocals. Maybe they waited just in
|case Lamar dropped another🔥|im shocked tracklib took this long to put this out its just the al green track with looped vocals
|maybe they waited just in case lamar dropped another|
|Your existence is a light 1
|your existence is a light
|
|2:37 THE GOOSEBUMPS I HADDDDDDDDDDD
|the goosebumps i haddddddd
|
|Comment with date and time you came here
|comment with date and time you came here
|
|Naaaa I need to hear that last part with the guitar yooooo do Family Ties dawggggggg
|naaaa i need to hear that last part with the guitar yooooo do family ties dawggggggg
|
|Fuck da allegations this shit too hard 🔥🔥🔥🔥
|fuck da allegations this shit too hard
|
|I love you$$
|i love you
|
|J cole tracks should be removed TBH. \n\nHe didn't mean it...
|j cole tracks should be removed tbh he didnt mean it
|
|That's not j cole
|thats not j cole

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These samples reveal:

- variability in language,
- sentiment polarity,
- informal and non-standard grammar,
- genre-specific terminology.

This informed the design of the text-cleaning pipeline.

5. Architecture for Batch and Streaming Analysis

A key objective was to design an architecture capable of handling both large-scale static data and continuously evolving real-time data. The architecture integrates multiple layers of processing, storage, and visualisation.

5.1 Batch Architecture Overview

The batch pipeline consisted of:

1. Data ingestion layer:
 - YouTube API requests.
 - JSONL storage.
2. Pre-processing layer (PySpark):
 - Duplicate removal.
 - Language filtering.
 - Text normalisation.
 - Sentiment scoring.

3. Classification layer:
 - Genre inference.
 - Artist inference.
4. Analysis layer:
 - Aggregations.
 - Average sentiment computation.
 - Cross-genre comparisons.
5. Visualisation layer:
 - Bar charts.
 - Line graphs.
 - Word clouds.

5.2 Streaming Architecture Overview

The streaming system captured Reddit comments in real time using PRAW and stored them in MongoDB. A continuous loop within Jupyter Notebook visualised rolling sentiment.

Components:

1. Data ingestion (PRAW):
 - Live stream of comments from three subreddits.
2. Storage (MongoDB):
 - Timestamped, structured records.
3. Processing layer:
 - Sentiment computed on retrieval.
 - Data filtered by subreddit.
4. Dashboard layer (matplotlib + ipywidgets):
 - Rolling sentiment curves.
 - Interactive artist comparison.
 - Word clouds.

5.3 Rationale for Architecture Choices

PySpark was chosen for batch processing due to its scalability and optimisation for distributed data.

MongoDB was selected for streaming storage because of its flexibility, schema-free design, and speed.

PRAW provides stable and well-documented access to Reddit's API.

Jupyter Notebook supports iterative development and visualisation within a single environment.

Matplotlib and Ipywidgets enable user-friendly dashboards without building a full web application.

6. Technologies and Methodologies Used

The project required integrating several technologies, each contributing to a distinct system capability.

6.1 Python Libraries

- PRAW: Reddit streaming.
- Google API Client: YouTube comment retrieval.
- TextBlob and VADER: sentiment analysis.
- Matplotlib: plotting and dashboards.
- Ipywidgets: interaction and filtering.
- PySpark: distributed processing.

6.2 Spark Pre-processing

PySpark was used extensively for:

- Filtering English comments using a stopwords-ratio heuristic.
- Removing duplicates based on comment_id or text/video pair.
- Creating user-defined functions (UDFs) for cleaning and scoring text.
- Aggregating sentiment by genre and artist.
- Converting Spark DataFrames to Pandas for visualisation.

6.3 Real-Time Data Handling

MongoDB acted as a buffer to hold continuously arriving Reddit comments. This enabled continuous updating of sentiment curves, safe storage in case of notebook or kernel failure, and efficient querying of new comments using timestamp filters.

6.4 Visualisation and Dashboard Methods

The dashboard incorporated:

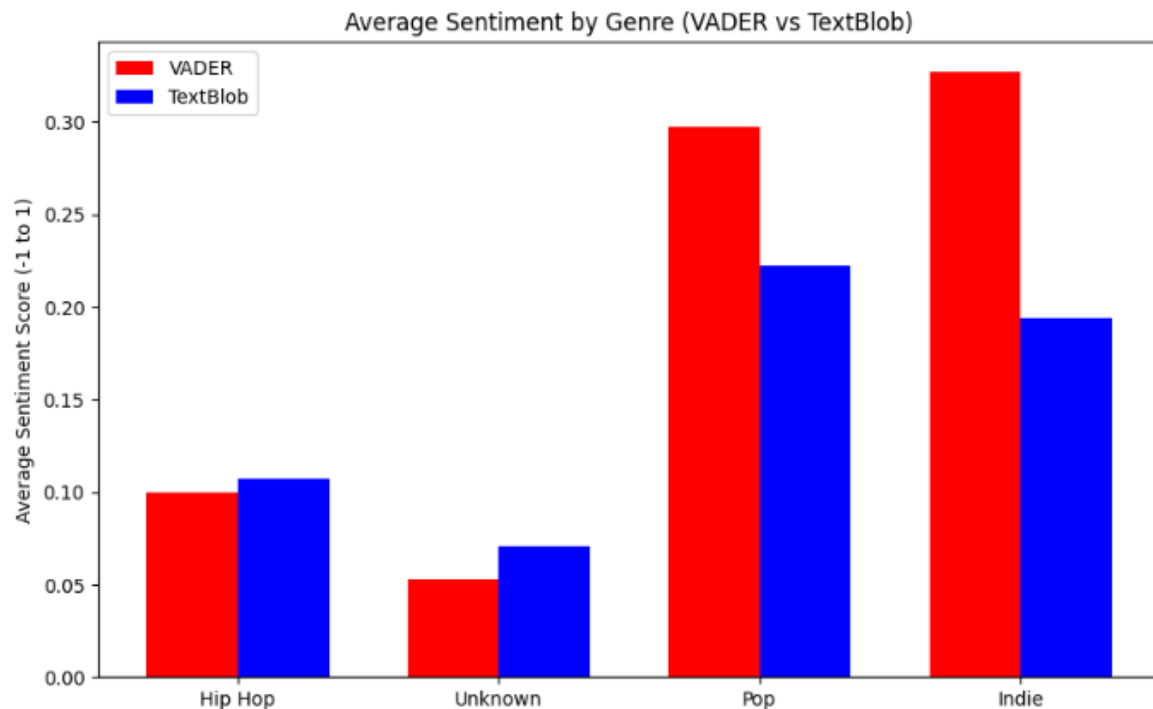
- Dropdown menus for selecting artists and sentiment models.
- Real-time updates every five seconds.
- Word cloud generation using cleaned text.
- Rolling averages to smooth noisy sentiment curves.

This created a simple but effective analytical interface.

7. Batch Results and Discussion

The batch analysis provides high-level insights into user sentiment and differences between genres and artists.

7.1 Genre Sentiment Results

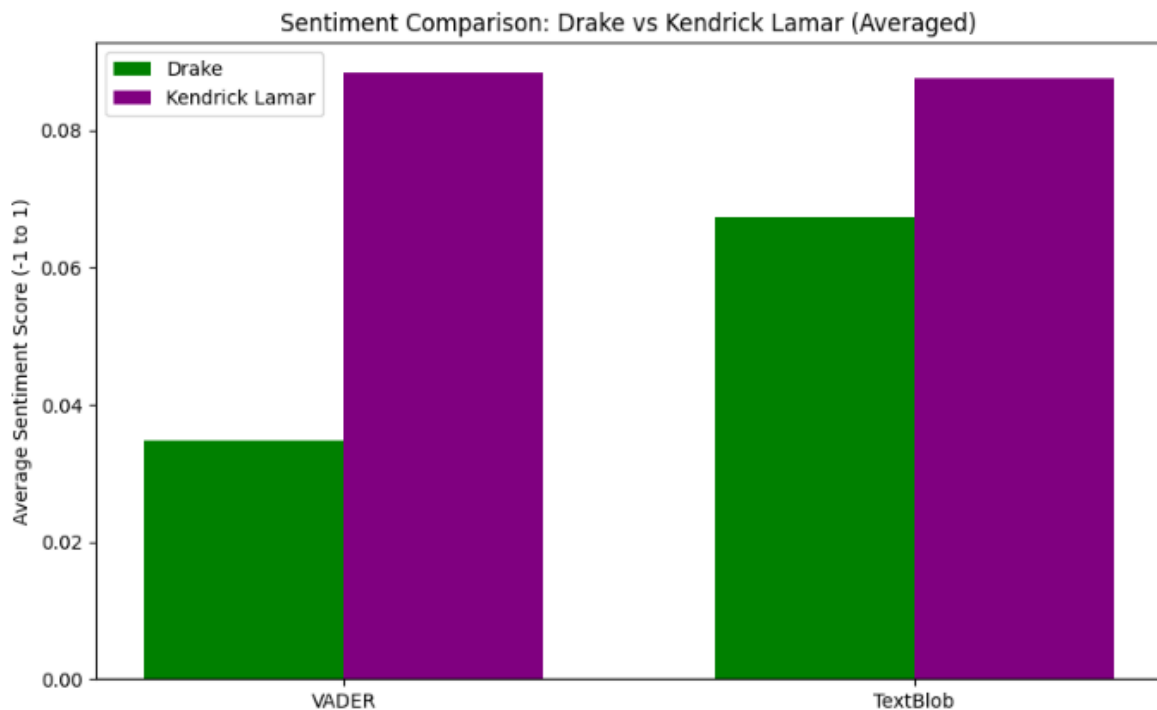


Key results:

- Indie comments demonstrated the highest positivity in VADER.
- Pop had the highest positivity in TextBlob, with very similar results to the Indie community.
- Hip Hop comments exhibited a much lower average positivity.

This aligns with known cultural dynamics. Hip Hop communities often engage in competitive or evaluative discourse, which may lead to more negative or polarised sentiment. Indie communities frequently celebrate emotional authenticity, leading to more positive language. Pop discussions involve celebrity culture, which can mix devotion with backlash.

7.2 Artist Sentiment Results



Despite sample size constraints caused by API limitations, several trends emerged:

- Drake commentary was more polarised, containing both strong praise and strong criticism.
- Kendrick Lamar commentary skewed consistently positive, with fewer negative remarks.

This supports the view that Drake, as a mainstream figure frequently involved in controversy, elicits strong emotional reactions. Kendrick Lamar's reputation as a critically respected artist may contribute to more stable sentiment outcomes.

7.3 Word Cloud Analysis



The Hip Hop word cloud contains strong language around performance, conflict, and lyrical evaluation.



The Indie word cloud reveals emotional, introspective vocabulary.

8. Streaming Results and Discussion

The streaming component provided live insight into community sentiment, revealing patterns unseen in batch data.

8.1 Real-Time Sentiment Trends

Real-Time Sentiment by Subreddit (VADER + TextBlob)

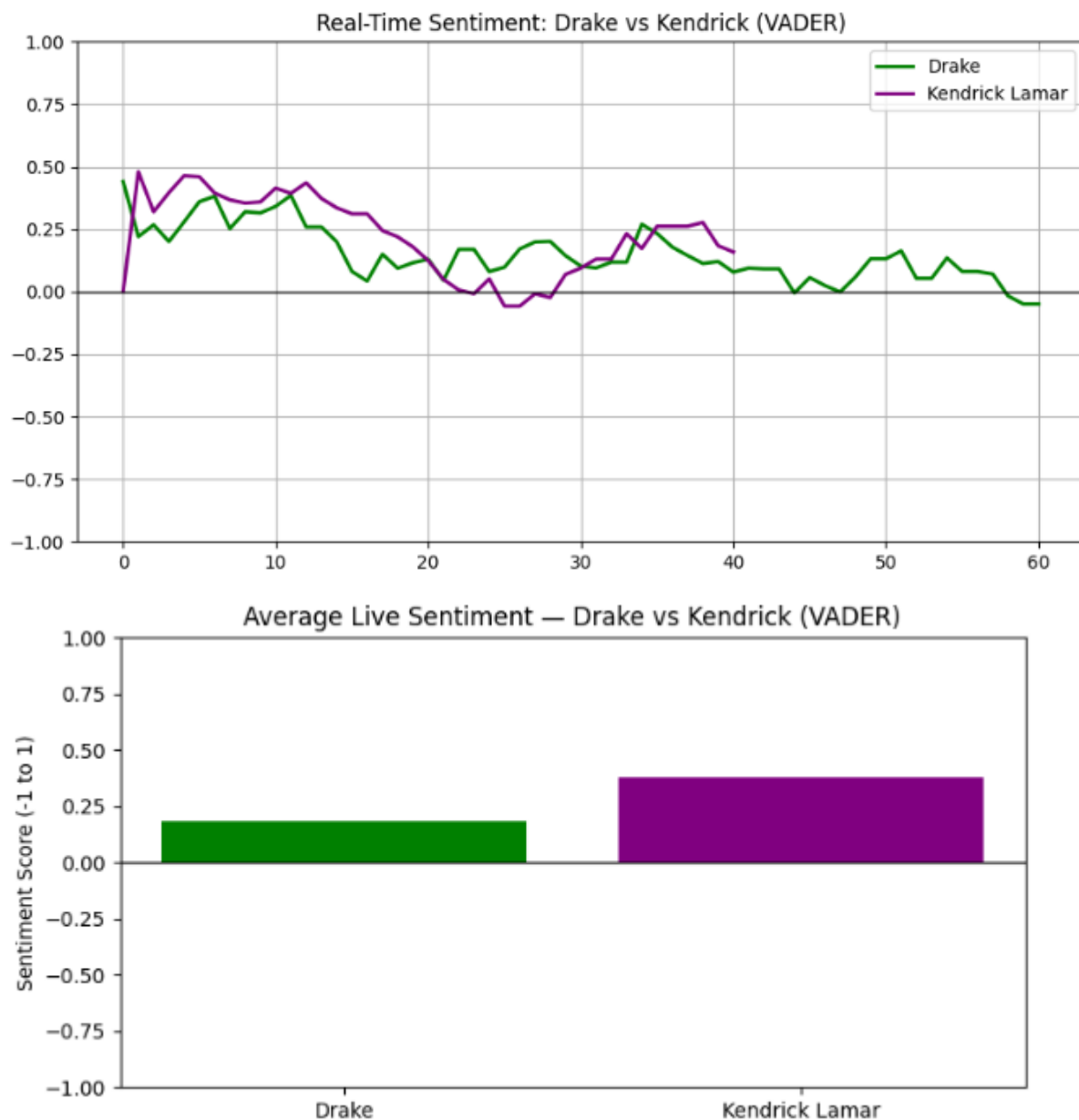


Key findings:

- HipHopHeads sentiment varied a bit and was mostly positive.
- IndieHeads remained largely positive, matching expectations from batch analysis.
- PopHeads exhibited moderate fluctuation, also remaining largely positive, matching batch analysis.

The contrast between genres highlights how temporal dynamics shape online communities.

8.2 Artist Dashboard Insights



Real-time artist analysis showed that Drake-related sentiment fluctuated more dramatically, consistent with batch findings. Kendrick Lamar-related sentiment was less frequent but mostly positive. This suggests differences in fan behaviour as well as the different public images of the artists.

8.3 Dashboard Functionality and Observations

The dashboard allowed real-time monitoring, interactive selection of sentiment models, word cloud generation for streaming text, and immediate visualisation of community mood.

Streaming visualisation captured sudden upward sentiment spikes, negative dips and temporary sentiment drops.

9. Ethical Considerations

9.1 Privacy and Consent

Although the batch and streaming pipelines initially ingested metadata fields such as the author's username (as supplied by Reddit and YouTube APIs), this information was used exclusively for internal processing tasks such as deduplication and filtering. Usernames were not used for profiling individuals, were not included in any visualisations or analyses, and were completely excluded from the written report.

9.2 Representativeness

Online communities do not represent the broader population. Music fans active on Reddit tend to be young, male, English-speaking and highly engaged.

Therefore, findings cannot be generalised beyond these groups.

9.3 Algorithmic Bias

YouTube's search algorithm influenced which videos were included, meaning data may reflect trending content over niche content, and highly engaged communities are overrepresented.

Reddit's upvote system similarly biases which comments appear prominently.

9.4 Linguistic and Cultural Bias

Sentiment tools are known to struggle with sarcasm, Hip Hop slang, African American Vernacular English (AAVE), Emojis and modern expressions.

These biases can distort sentiment scores, particularly for Hip Hop.

9.5 Researcher Responsibility

To maintain ethical integrity, results are presented with caveats and limitations are acknowledged.

10. Limitations

The limitations of the study include:

- Restricted API quota limiting batch data volume.
- Sentiment models that lack contextual or cultural awareness.
- Temporal bias in streaming data.
- Classification heuristics that are imperfect.
- Platform-driven distortions in data visibility.
- Dashboard limitations due to Jupyter Notebook constraints.

These limitations informed a cautious interpretive stance.

11. Conclusion

This project demonstrates how batch and real-time sentiment analysis can be integrated to provide a multidimensional view of online music communities. Batch analysis showed clear genre differences and highlighted contrast between Drake and Kendrick Lamar's sentiment profiles. Streaming analysis revealed temporal patterns and community volatility.

Together, these findings highlight how music fandoms differ not only in content but also in emotional expression, stability, and responsiveness to cultural events. This work provides a foundation for deeper computational and cultural exploration of music communities and illustrates the methodological value of combining static and dynamic data analysis.

Future work could incorporate:

- Transformer-based NLP (e.g., BERT) for contextual analysis.
- Larger streaming windows.
- Cross-platform triangulation (Twitter, TikTok).
- Advanced visualisation dashboards (PowerBI, Tableau).

Ultimately, the study contributes to understanding how music communities express emotion online and how computational tools can support this analysis.